

9	(a)	The magnetic flux linkage is defined as the product of the number of turns of the coil and the component of magnetic flux density passing perpendicularly through a cross sectional area of the coil.		B1 B1
	(b)	(i)	Magnetic flux density at the centre of the solenoid, $B = \mu_0 n I = (4\pi \times 10^{-7}) \left(\frac{900}{30.0 \times 10^{-2}} \right) (2.0)$ $= 7.54 \times 10^{-3} \text{ T}$ Magnetic flux through flat coil = magnetic flux through solenoid $\phi = B A = (7.54 \times 10^{-3}) (3.2 \times 10^{-4}) = 2.41 \times 10^{-6} \text{ Wb}$	M1 M1 A0
		(ii)1.	Magnitude of e.m.f. $\varepsilon = N \frac{d\phi}{dt} \approx N \frac{\Delta\phi}{\Delta t} = (250) \left(\frac{2.41 \times 10^{-6}}{0.45} \right)$ $= 1.338 \times 10^{-3} = 1.34 \times 10^{-3} \text{ V}$	M1 A1
		(ii)2.	charge = current x time $= \frac{\varepsilon}{R} \times t = \left(\frac{1.338 \times 10^{-3}}{35.0} \right) (0.45)$ $= 1.722 \times 10^{-5} = 1.72 \times 10^{-5} \text{ C}$	M1 A1
		(ii)3.	charge = (average current) x (time) $= \left(\frac{1}{2} I \right) (t) = \left(\frac{2.0}{2} \right) (0.45)$ $= 0.45 \text{ C}$	M1 A1
	(c)	(i)	$evB = eE$ $B = \frac{E}{v} = \frac{\Delta V}{dv} = \frac{(15 - (-15))}{0.040 (4.5 \times 10^6)}$ $= 1.6667 \times 10^{-4} = 1.67 \times 10^{-4} \text{ T}$	M1 A1
		(ii)	The electrons experienced a constant magnetic force always perpendicular to its direction of motion/velocity. This force does zero work on the electrons hence the speed remains constant. Hence the magnetic force provides the centripetal force for circular motion.	B1 B1
		(iii)	The centripetal force is provided by the magnetic force. $evB = \frac{mv^2}{r}$ $r = \frac{mv^2}{evB} = \frac{mv}{eB} = \frac{(9.11 \times 10^{-31}) (4.50 \times 10^6)}{(1.6 \times 10^{-19}) (0.50 \times 10^{-3})}$ $= 0.05124 = 0.0512 \text{ m}$	M1 A1

		(iv)	$evB = eE$ $v = \frac{E}{B} = \frac{7.50 \times 10^2}{1.6667 \times 10^{-4}} = 4.49991 \times 10^6 = 4.50 \times 10^6 \text{ ms}^{-1}$ Accept if candidates just state the speed, realising that the speed of protons is the same as that of the electrons passing through the same velocity selector.	A1
		(v)	The diagram shows a beam of electrons entering a region between two parallel plates. The top plate is at +15 V and the bottom plate is at -15 V. The distance between the plates is 15.0 cm. The beam enters a shaded rectangular region labeled 'second region of magnetic field' which has a width of 12.0 cm and a height of 30.0 cm. Inside this region, protons curve upwards and exit on the right, while electrons curve downwards in a semi-circular path and exit on the left.	
			circular path of electrons in clockwise direction (semi-circle) Relative size of radius and proton must exit on right side of magnetic field Protons curve upwards in anti-clockwise direction	A1 A1 A1
			Total:	20